

Operating Systems Laboratory

MacBook Student Guide: Ubuntu VM + GCC

Complete Ubuntu installation using UTM, architecture selection, setup, GCC and troubleshooting

Important: macOS does not use WSL. If Ubuntu/Linux is not already installed, create an Ubuntu virtual machine. This guide uses UTM because it provides a practical VM workflow for macOS. If Ubuntu is already installed in another VM such as Parallels or VMware, the student can skip to the Ubuntu verification and GCC sections.

MacBook type	Ubuntu image
Apple Silicon (M1/M2/M3/M4 etc.)	ARM64 / ARM 64-bit Ubuntu image
Intel Mac	Intel/AMD 64-bit (x86_64/AMD64) Ubuntu image
Virtualization app in this guide	UTM
Compiler	GCC inside Ubuntu
Working folder	~/OS

1. First Check: Is Ubuntu Already Installed?

Press Command + Space and search for UTM, Parallels Desktop, VMware Fusion, VirtualBox or Ubuntu. If a virtual machine named Ubuntu already exists, open that VM and start it. If no Ubuntu/Linux VM exists, continue with this guide.

2. Identify the Mac Architecture

Apple menu  → About This Mac. If the Mac shows an Apple M-series chip, use ARM64 Ubuntu. If it shows an Intel processor, use the Intel/AMD 64-bit Ubuntu image.

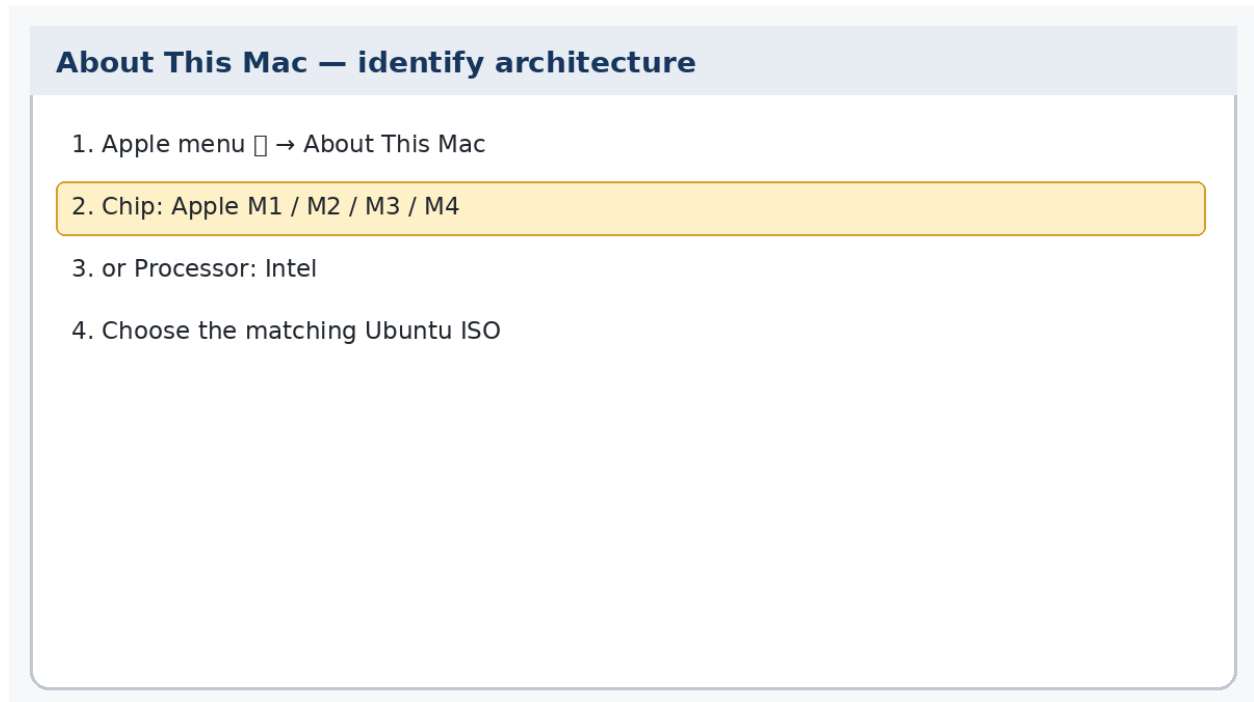


Figure: Checking whether the Mac is Apple Silicon or Intel. The exact UI may vary slightly by Windows/macOS/UTM version.

3. Back Up Important Mac Files

A virtual machine normally stores its data as a VM file on the Mac. Before changing VM storage or deleting an existing VM, back up important student files. Do not delete a VM simply because Ubuntu fails to boot; troubleshoot it first.

4. Install UTM

Download UTM from its official website: <https://utmapp.io/> . Install it like a normal macOS application and open it.

5. Download Ubuntu

Use the official Ubuntu Desktop download page. Select ARM 64-bit for Apple Silicon or Intel/AMD 64-bit for an Intel Mac. Ubuntu publishes separate architecture images.

Download Ubuntu

1. Open ubuntu.com/download/desktop
2. Apple Silicon → ARM64
3. Intel Mac → Intel/AMD 64-bit
4. Download the ISO

Figure: Selecting the Ubuntu ISO architecture. The exact UI may vary slightly by Windows/macOS/UTM version.

6. Create the Ubuntu Virtual Machine in UTM

Open UTM → click + → choose Virtualize → choose Linux. UTM's Ubuntu guide follows this general workflow: create a Linux VM, select the Ubuntu ISO, allocate CPU/RAM/disk, save the VM and start it.

UTM — create Linux VM

1. Open UTM
2. Click +
3. Select Virtualize
4. Select Linux
5. Continue

Figure: UTM Linux virtual machine creation wizard. The exact UI may vary slightly by Windows/macOS/UTM version.

7. Select the Ubuntu ISO

Click Browse and select the Ubuntu .iso file downloaded from the official Ubuntu website. Double-check that the architecture matches the Mac.

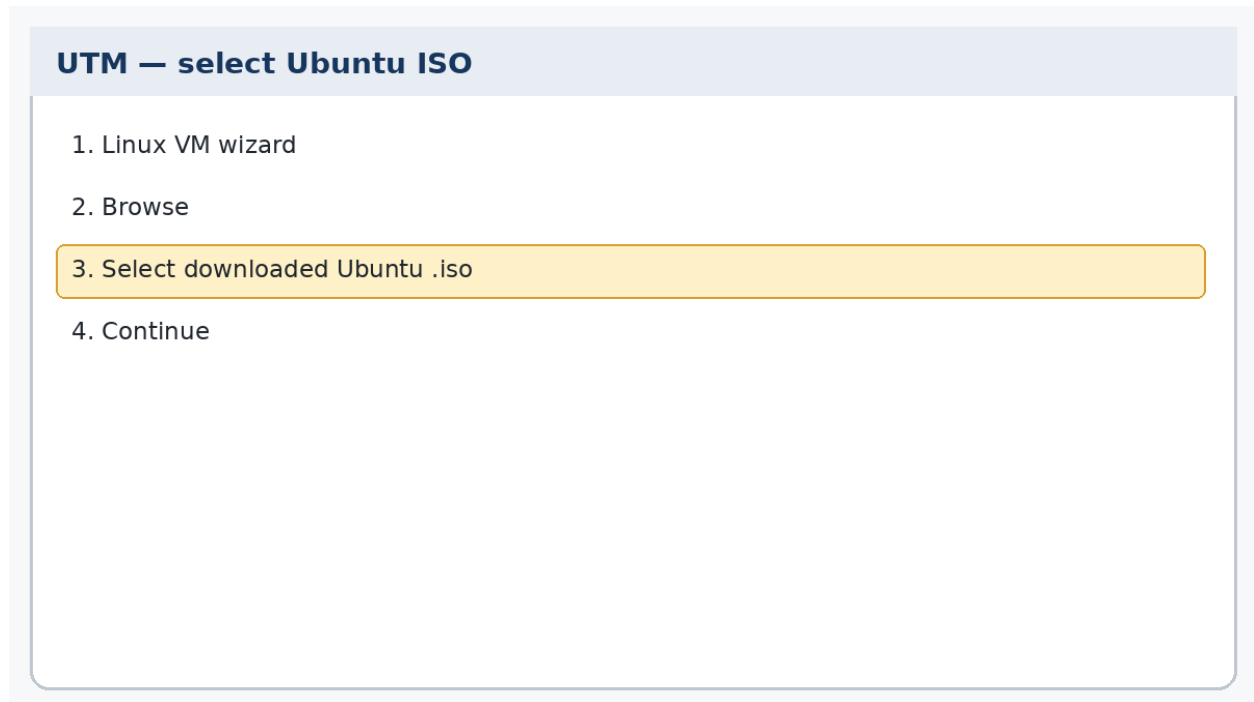


Figure: Selecting the Ubuntu ISO in UTM. The exact UI may vary slightly by Windows/macOS/UTM version.

8. Recommended VM Resources for OS Lab

Use these as starting values; adjust if the Mac has less RAM/storage:

Resource	Recommended starting point
CPU	2–4 virtual cores
RAM	4 GB minimum; 6–8 GB preferred if the Mac has enough memory
Disk	30–50 GB
Network	Shared/NAT network enabled
Shared folder	Optional; use only after Ubuntu is working

Do not allocate all Mac RAM/CPU to Ubuntu. Keep enough resources for macOS.

UTM — VM resources

1. CPU: 2–4 cores
2. RAM: 4–8 GB
3. Disk: 30–50 GB
4. Shared directory: optional
5. Save → Run

Figure: Example UTM resource configuration. The exact UI may vary slightly by Windows/macOS/UTM version.

9. Save and Start the VM

Click Save, select the new Ubuntu VM, and click Run/Play. Ubuntu should boot from the ISO and display its installer.

10. Ubuntu Installation Wizard

Follow the installer: choose language → keyboard → network → installation type → create username/password → complete installation. Keep the default choices unless your course instructor specifies otherwise.

Ubuntu installer

1. Choose language
2. Keyboard layout
3. Connect to network
4. Install Ubuntu
5. Create username and password

Figure: Ubuntu installation wizard sequence. The exact UI may vary slightly by Windows/macOS/UTM version.

11. First Ubuntu Boot

After installation, restart the VM. If UTM asks whether to remove/unmount the installer ISO, remove it so the VM boots from the virtual disk. Log in with the Ubuntu account created during installation.

Open Applications → Terminal. For a terminal-only OS lab, the terminal is the main tool students need.

Ubuntu first boot

1. Ubuntu desktop / login screen
2. Log in with your Ubuntu account
3. Open Applications → Terminal

Figure: Opening Terminal inside Ubuntu. The exact UI may vary slightly by Windows/macOS/UTM version.

12. Verify Ubuntu

```
lsb_release -a
uname -m
pwd
```

For Apple Silicon Ubuntu, `uname -m` will normally report an ARM64 architecture such as `aarch64`. Intel Ubuntu normally reports `x86_64`.

13. Update Ubuntu

```
sudo apt update
```

Enter the Ubuntu password. Password characters will not be displayed while typing.

14. Install GCC

```
sudo apt install build-essential
```

Then verify:

```
gcc --version
make --version
```

Install GCC in Ubuntu

1. `sudo apt update`

2. `sudo apt install build-essential`

3. `gcc --version`

Figure: Installing GCC and build tools inside Ubuntu. The exact UI may vary slightly by Windows/macOS/UTM version.

15. Create the OS Laboratory Folder

```
mkdir -p ~/OS  
cd ~/OS  
pwd
```

Keep C source files here. This avoids permission problems associated with shared macOS folders or system directories.

Create OS lab workspace

1. mkdir ~/OS
2. cd ~/OS
3. pwd
4. /home/student/OS

Figure: Creating a dedicated OS laboratory folder. The exact UI may vary slightly by Windows/macOS/UTM version.

16. Test GCC with a C Program

```
nano hello.c
```

Enter:

```
#include <stdio.h>
```

```
int main()  
{  
    printf("Hello from Ubuntu on Mac\n");  
    return 0;  
}
```

Save with Ctrl+O → Enter; exit with Ctrl+X. Compile and run:

```
gcc hello.c -o hello  
./hello
```


Compile and run C program

1. `gcc hello.c -o hello`
2. `./hello`
3. Hello from Ubuntu on Mac

Figure: Compiling and executing a C program inside Ubuntu on Mac. The exact UI may vary slightly by Windows/macOS/UTM version.

17. Important: MacBook Students Do NOT Need WSL

WSL is a Windows feature. A Mac student should not search for or install WSL. The equivalent classroom workflow is:

macOS → UTM virtual machine → Ubuntu → Terminal → GCC → OS programs

If Ubuntu is already installed in Parallels, VMware, VirtualBox or another working VM, use that existing Ubuntu environment instead of installing a second one.

18. If Ubuntu Is Already Installed but Won't Start

- Open the VM application and check whether the Ubuntu VM is present.
- Do not delete the VM.
- Check that the Ubuntu virtual disk is attached.
- If the VM boots into EFI/UEFI shell, check that the Ubuntu installer ISO or installed virtual disk is correctly attached.
- If the problem began after an update, restart the VM and check UTM/VM settings.

19. UTM: Cannot Boot the Ubuntu Installer

If UTM shows an EFI shell or cannot find the installer, check the VM's CD/DVD/ISO setting and make sure the Ubuntu ISO is mounted. UTM's Ubuntu documentation specifically recommends checking that the installer ISO is selected in the VM's removable drive configuration.

UTM troubleshooting

1. EFI shell / cannot boot → check ISO is mounted
2. No network → verify VM network adapter
3. Black screen → restart VM / check UTM settings
4. Permission denied on shared folder → use Linux home first

Figure: Common UTM troubleshooting points. The exact UI may vary slightly by Windows/macOS/UTM version.

20. UTM: No Internet Inside Ubuntu

Check that the VM has a network device enabled. In Ubuntu, run:

```
ip link  
ip addr
```

Then test connectivity:

```
ping -c 4 archive.ubuntu.com
```

If networking is unavailable, stop the VM, check its network settings, restart it, and test again. UTM documents network-adaptor ordering/configuration issues on some Linux VM setups.

21. Permission Denied in a Shared macOS Folder

Do not begin by changing permissions recursively. First work in the Ubuntu home directory:

```
cd ~  
mkdir -p ~/OS  
cd ~/OS
```

If a UTM shared directory is required, UTM documents that UID/GID differences between macOS and Linux can cause access-denied errors. For the OS lab, the simplest and safest choice is to keep source code in ~/OS unless file sharing is specifically required.

22. GCC Installation Problems

Problem	Solution
gcc: command not found	sudo apt update → sudo apt install build-essential
Unable to locate package	Check internet; run sudo apt update again
Permission denied	Use sudo for apt installation; work in ~/OS
apt/dpkg lock	Wait for another package operation to finish; do not delete lock files blindly
Compilation error	Read the first compiler error carefully; it is usually the most useful one

23. MacBook Architecture and C Programs

Your normal C programs for OS lab — including `fork()`, `exec()`, process IDs, pipes and scheduling algorithms — can be compiled and executed inside Ubuntu. The exact CPU architecture may differ between Intel and Apple Silicon, but the Linux programming environment remains Ubuntu.

24. Student Verification Checklist

- Mac architecture identified (Apple Silicon or Intel)
- UTM/VM software installed or existing VM located
- Correct Ubuntu architecture ISO downloaded
- Ubuntu VM created
- Ubuntu installation completed
- Ubuntu boots from virtual disk
- Terminal opens
- `lsb_release -a` works
- `uname -m` checked
- `sudo apt update` works
- `build-essential` installed
- `gcc --version` works
- `~/OS` created
- `hello.c` compiles
- `./hello` executes successfully

25. Final Commands for the OS Laboratory

```
cd ~/OS
sudo apt update
sudo apt install build-essential
gcc --version
gcc program.c -o program
./program
```

Once these commands work, the MacBook student is ready for the Operating Systems laboratory C programs.

Official References

Official reference: Ubuntu Desktop downloads: <https://ubuntu.com/download/desktop>

Official reference: Ubuntu Desktop installation documentation:

<https://documentation.ubuntu.com/desktop/en/latest/tutorial/install-ubuntu-desktop/>

Official reference: UTM Ubuntu guide: <https://docs.getutm.app/guides/ubuntu/>

Official reference: UTM Linux guest support: <https://docs.getutm.app/guest-support/linux/>

Note: Ubuntu's current desktop download page provides both Intel/AMD 64-bit and ARM 64-bit images. The exact installer screens and UTM menus can change between releases, so students should follow the closest matching screen shown in this guide.